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MAGNETIC DEVICE

Abstract

A magnetic device includes a first element portion including a first magnetic layer, a first non-magnetic member, a first magnetic member, and a first intermediate layer. The first non-magnetic member includes at least one of first and second materials. The first material includes first and second elements. The first element includes one of first and second type elements. The second element includes at least one element selected from the group consisting of oxygen and nitrogen. The first type element includes at least one element selected from the group consisting of Ru, Ta, Mo, W, Hf, Cr, Cu, Pd, V, Ti, and Zn. The second type element includes at least one selected from the group consisting of Mg and Al. The second material includes a third element and a fourth element. The third element includes at least one selected from the group consisting of Pt, Cu, and Hf.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation application of International Application PCT/JP2022/044174, filed on Nov. 30, 2022; the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to a magnetic device.

BACKGROUND

[0003] Magnetic devices including a magnetic layer are used in a variety of applications. Stable operation is desired in the magnetic devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is a schematic cross-sectional view illustrating a magnetic device according to a first embodiment;

[0005] FIG. 2 is a schematic perspective view illustrating the magnetic device according to the first embodiment;

[0006] FIG. 3 is a schematic cross-sectional view illustrating a magnetic device according to the first embodiment;

[0007] FIGS. 4A and 4B are schematic cross-sectional views illustrating a magnetic device according to the first embodiment;

[0008] FIGS. 5A and 5B are schematic cross-sectional views illustrating a magnetic device according to the first embodiment;

[0009] FIGS. 6A and 6B are schematic cross-sectional views illustrating a magnetic device according to the first embodiment;

[0010] FIGS. 7A and 7B are schematic cross-sectional views illustrating a magnetic device according to the first embodiment;

[0011] FIG. 8 is a schematic cross-sectional view illustrating a magnetic device according to a second embodiment; and

[0012] FIG. 9 is a schematic cross-sectional view illustrating a magnetic device according to a third embodiment.

DETAILED DESCRIPTION

[0013] According to one embodiment, a magnetic device includes a first element portion. The first element portion includes a first magnetic layer, a first non-magnetic member, a first magnetic member, and a first intermediate layer. The first non-magnetic member is conductive. A direction from the first magnetic layer to the first non-magnetic member is along a first direction. The first magnetic member is provided between the first magnetic layer and the first non-magnetic member, and is in contact with the first non-magnetic member. The first intermediate layer is non-magnetic and is provided between the first magnetic layer and the first magnetic member. The first non-magnetic member includes at least one of a first material or a second material. The first material includes a first element and a second element. The first element includes one of a first type element and a second type element. The second element includes at least one element selected from the group consisting of oxygen and nitrogen. The first type element includes at least one element

selected from the group consisting of Ru, Ta, Mo, W, Hf, Cr, Cu, Pd, V, Ti, and Zn. The second type element includes at least one selected from the group consisting of Mg and Al. The second material includes a third element and a fourth element. The third element includes at least one selected from the group consisting of Pt, Cu, and Hf, the fourth element including Al.

[0014] Various embodiments are described below with reference to the accompanying drawings.

[0015] The drawings are schematic and conceptual; and the relationships between the thickness and width of portions, the proportions of sizes among portions, etc., are not necessarily the same as the actual values. The dimensions and proportions may be illustrated differently among drawings, even for identical portions.

[0016] In the specification and drawings, components similar to those described previously or illustrated in an antecedent drawing are marked with like reference numerals, and a detailed description is omitted as appropriate.

First Embodiment

[0017] FIG. **1** is a schematic cross-sectional view illustrating a magnetic device according to a first embodiment.

[0018] FIG. **2** is a schematic perspective view illustrating the magnetic device according to the first embodiment.

[0019] A magnetic device **110** according to the embodiment includes a first element portion **10E**. The first element portion **10E** includes a first magnetic layer **11**, a first non-magnetic member **31**, a first magnetic member **21**, and a first intermediate layer **15**.

[0020] The first non-magnetic member **31** is conductive. A direction from the first magnetic layer **11** to the first non-magnetic member **31** is along a first direction **D1**.

[0021] The first direction **D1** is defined as a Z-axis direction. One direction perpendicular to the Z-axis direction is defined as an X-axis direction. A direction perpendicular to the Z-axis and X-axis directions is defined as a Y-axis direction.

[0022] The first magnetic member **21** is provided between the first magnetic layer **11** and the first non-magnetic member **31**. The first magnetic member **21** is in contact with the first non-magnetic member **31**. The first intermediate layer **15** is provided between the first magnetic layer **11** and the first magnetic member **21**. The first intermediate layer **15** is non-magnetic. For example, the first intermediate layer **15** may be in contact with the first magnetic layer **11** and the first magnetic member **21**.

[0023] The first non-magnetic member **31** includes at least one of a first material or a second material. The first material includes a first element including one of a first type element and a second type element, and a second element including at least one selected from the group consisting of oxygen and nitrogen. The first type element includes at least one selected from the group consisting of Ru, Ta, Mo, W, Hf, Cr, Cu, Pd, V, Ti, and Zn. The second type element includes at least one selected from the group consisting of Mg and Al. The second material includes a third element including at least one selected from the group consisting of Pt, Cu, and Hf, and a fourth element including Al. For example, the first non-magnetic member **31** includes Ru and oxygen. For example, the first non-magnetic member **31** includes Al, Pt, etc. This provides stable operation, as described below. Examples of the first material and the second material will be described later.

[0024] In the embodiment, the first intermediate layer **15** includes, for example, at least one selected from the group consisting of MgO, CaO, SrO, TiO, VO, NbO, and Al.sub.2O.sub.3. The first element portion **10E** is, for example, a TMR (Tunnel Magneto Resistance) element.

[0025] The first magnetic layer **11** is, for example, a magnetization free layer. The first magnetic member **21** is, for example, a magnetization reference layer. The magnetization of the first magnetic member **21** is less likely to change than the magnetization of the first magnetic layer **11**.

[0026] The first magnetic layer **11** includes at least one selected from the group consisting of Fe, Co, and Ni. The first magnetic layer **11** may further include boron. The first magnetic layer **11** is, for example, a ferromagnetic layer.

[0027] As shown in FIG. 1, in one example, the first magnetic member **21** includes a plurality of first magnetic films **21m** and a plurality of first non-magnetic films **21n**. In the first direction **D1**, one of the plurality of first non-magnetic films **21n** is located between one of the plurality of first magnetic films **21m** and another one of the plurality of first magnetic films **21m**. One of the plurality of first magnetic films **21m** is located between one of the plurality of first non-magnetic films **21n** and another one of the plurality of first non-magnetic films **21n**. For example, the first magnetic films **21m** and the first non-magnetic films **21n** are alternately provided. The first magnetic film **21m** contacts the first non-magnetic film **21n**.

[0028] For example, the first magnetic member **21** may be a synthetic anti-ferromagnetic (SAF) layer. One of the plurality of first magnetic films **21m** may be antiferromagnetically coupled to another one of the plurality of first magnetic films **21m**. For example, the SAF structure stabilizes the magnetization of the plurality of first magnetic films **21m**. The magnetization of the first magnetic member **21** is stabilized.

[0029] For example, the first magnetic member **21** may satisfy the following first condition or second condition. In the first condition, one of the plurality of first non-magnetic films **21n** includes Ru. The thickness **t21n** of one of the plurality of first non-magnetic films **21n** along the first direction **D1** is not less than 0.2 nm and not more than 2 nm.

[0030] In the second condition, one of the plurality of first non-magnetic films **21n** includes Ir. The thickness **t21n** of one of the plurality of first non-magnetic films **21n** along the first direction **D1** is not less than 0.2 nm and not more than 2 nm.

[0031] Such first condition or second condition makes it easier to obtain antiferromagnetic coupling. Meanwhile, the plurality of first magnetic films **21m** include at least one selected from the group consisting of Fe, Co, and Ni. The plurality of first magnetic films **21m** may further include boron. The thickness **t21m** of the plurality of first magnetic films **21m** along the first direction **D1** may be, for example, not less than 0.2 nm and not more than 5 nm.

[0032] As already explained, in one example, the first magnetic member **21** functions as a magnetization reference layer. In a reference example, an antiferromagnetic member (e.g., IrMn or PtMn) is provided on the first magnetic member **21**. This allows the magnetization of the first magnetic member **21** to be fixed in a desired direction. In this reference example, it was found that the magnetization of the first magnetic member **21** is likely to become unstable when the size of the magnetic element (e.g., the first element portion **10E**) becomes smaller. For example, when the size of the magnetic element (e.g., the first element portion **10E**) becomes smaller, the size of the magnetic element approaches the size of the crystal grains included in the antiferromagnetic member. This causes the state of the crystal grains included in the antiferromagnetic member in the magnetic element to become non-uniform. As a result, it is considered that the magnetization of the first magnetic member **21** becomes unstable.

[0033] In contrast, in the embodiment, the first non-magnetic member **31** above-mentioned is provided instead of the antiferromagnetic member. The first non-magnetic member **31** includes the above-mentioned first material or second material. The first material includes the first element and the second element. For example, it is considered that the volume of the first non-magnetic member **31** changes when the second element (oxygen and/or nitrogen) is introduced into a member including the first element. As a result, stress is generated in the first non-magnetic member **31**. The second material includes the third element and the fourth element (Al). For example, it is considered that the volume of the first non-magnetic member **31** changes when the third element (Pt, Cu, and/or Hf, etc.) is introduced into a member including the fourth element (Al). As a result, stress is generated in the first non-magnetic member **31**. It is considered that the stress is applied to the first magnetic member **21**, and the magnetization direction of the first magnetic member **21** is controlled.

[0034] In the first non-magnetic member **31**, the above-mentioned crystal grain problem in the antiferromagnetic member does not occur. Even if the size of the magnetic element (for example,

the first element portion **10E**) is reduced, the magnetization of the first magnetic member **21** is stably controlled.

[0035] By stabilizing the magnetization of the first magnetic member **21**, the characteristics of the magnetic element are stabilized. For example, the stability of the characteristics of the first magnetic member **21** is improved. For example, a large resistance change rate is obtained. For example, a large read signal is obtained. For example, stress is applied to the first magnetic layer **11**. For example, good retention is obtained. Stable characteristics can be maintained even if the size of the first element portion **10E** is reduced. According to the embodiment, a magnetic device capable of stable operation can be provided.

[0036] In the embodiment, for example, the first non-magnetic member **31** can apply stress to the first magnetic member **21**. The stress may be, for example, one of tensile stress and compressive stress. The stress may have a component in a direction (a direction along the X-Y plane) that crosses the first direction **D1**.

[0037] As described above, the first material includes a first element including one of a first type element and a second type element, and a second element including at least one selected from the group consisting of oxygen and nitrogen. The first type element includes at least one selected from the group consisting of Ru, Ta, Mo, W, Hf, Cr, Cu, Pd, V, Ti, and Zn. For example, when oxygen is introduced into a film of the first type element, the volume of the film increases. The increase in volume is considered to be due to, for example, the difference between the structure (e.g., crystal structure) of the film of the first type element and the structure (e.g., crystal structure) of the film including the first type element and oxygen.

[0038] The effect of the increase in volume is large for Ru, Ta, Mo, or W. The effect of the increase in volume is moderate for Hf, Cr, Cu, and Pd. The effect of the increase in volume is relatively small for V, Ti, or Zn.

[0039] The second type element includes at least one selected from the group consisting of Mg and Al. For example, when oxygen is introduced into a film of the second type element, the volume of the film decreases. The decrease in volume is thought to be due to, for example, the difference between the structure (e.g., crystal structure) of the film of the second type element and the structure (e.g., crystal structure) of the film including the second type element and oxygen.

[0040] Among the various elements mentioned above, Hf, Al, and Mg are difficult to mix with the layer including Co, Fe, etc. included in the first magnetic member **21**. In a case where the first non-magnetic member **31** includes Hf, Al, or Mg, stress can be applied to the first magnetic member **21** while suppressing adverse effects on the characteristics of the first magnetic member **21**.

[0041] As described above, when the first non-magnetic member **31** includes the first material or the second material, stress is generated between the first non-magnetic member **31** and the first magnetic member **21**. This stress may have anisotropy in the X-Y plane. The anisotropy may be induced by various configurations.

[0042] For example, as shown in FIG. 1 and FIG. 2, the magnetic device **110** may further include a first conductive member **51**. The first conductive member **51** includes a first conductive portion **51a**, a second conductive portion **51b**, and a third conductive portion **51c**. The third conductive portion **51c** is provided between the first conductive portion **51a** and the second conductive portion **51b**. A second direction **D2** from the first conductive portion **51a** to the second conductive portion **51b** crosses the first direction **D1**. The second direction **D2** is, for example, the X-axis direction.

[0043] As shown in FIG. 2, the first conductive portion **51a** and the second conductive portion **51b** are present on both sides of the position of the first element portion **10E** in the X-axis direction. On the other hand, the first conductive member **51** is not present on both sides of the position of the first element portion **10E** in the Y-axis direction. In this way, anisotropy exists in the X-Y plane. This may induce anisotropy in the stress. For example, the magnetization of the first magnetic member **21** may be controlled by the stress anisotropy.

[0044] In another example, as described below, an insulating member may be provided around the

first element portion **10E**, and the configuration of the insulating member may vary within the X-Y plane. This may induce anisotropy in the stress. For example, the magnetization of the first magnetic member **21** may be controlled by the stress anisotropy.

[0045] In further another example, anisotropy may be provided in the shape of at least one of the first non-magnetic member **31** or the first magnetic member **21**. This may induce anisotropy in the stress. For example, the magnetization of the first magnetic member **21** may be controlled by the stress anisotropy.

[0046] For example, as shown in FIG. 2, the length **31y** of the first non-magnetic member **31** along the third direction may be different from the length **31x** of the first non-magnetic member **31** along the second direction **D2**. The third direction **D3** crosses a plane including the first direction **D1** and the second direction **D2**. The third direction **D3** is, for example, the Y-axis direction.

[0047] In one example, the length **31y** is longer than the length **31x**. This results in in-plane anisotropy in the stress generated in the first non-magnetic member **31**.

[0048] For example, as shown in FIG. 2, the length **21y** of first magnetic member **21** along third direction **D3** may be different from the length **21x** of first magnetic member **21** along second direction **D2**. For example, the length **21y** is longer than the length **21x**. This causes in-plane anisotropy in the stress applied to first magnetic member **21**.

[0049] Due to the in-plane anisotropy of the stress applied to first magnetic member **21**, in-plane anisotropy in the lattice length of first magnetic member **21** may be generated.

[0050] For example, a third direction lattice length along the third direction **D3** of at least a part of the first magnetic member **21** may be different from a second direction lattice length along the second direction **D2** of at least a part of the first magnetic member **21**. For example, the third direction lattice length is longer than the second direction lattice length.

[0051] In a case where the third direction lattice length is longer than the second direction lattice length, the first magnetic member **21** is pulled in the third direction **D3**. At this time, the first magnetic member **21** is compressed in the second direction **D2**.

[0052] In a case where the third direction lattice length is shorter than the second direction lattice length, the first magnetic member **21** is pulled in the second direction **D2**. At this time, the first magnetic member **21** is compressed in the third direction **D3**.

[0053] In a case where the magnetostriction constant of the first magnetic member **21** is positive, the magnetization of the first magnetic member **21** is along the direction of tension. In a case where the magnetostriction of the first magnetic member **21** is negative, the magnetization of the first magnetic member **21** is along the direction of compression.

[0054] In the embodiment, for example, stress is applied to the first magnetic member **21** by the first non-magnetic member **31**. As described above, anisotropy of the stress may be generated by at least one of the shape of the first conductive member **51**, the configuration of the insulating member provided around the first element portion **10E**, the shape of the first non-magnetic member **31**, or the shape of the first magnetic member **21**. By generating anisotropic stress in the first magnetic member **21**, the magnetization of the first magnetic member **21** is stably controlled. Various examples of anisotropy induced in the stress will be described later.

[0055] Stress is introduced to the first magnetic member **21** due to the influence of the first non-magnetic member **31**. The stress may vary along the Z-axis direction depending on the distance from the first non-magnetic member **31**. The lattice length may vary along with the stress. For example, the lattice length of the first magnetic member **21** may vary along the Z-axis direction. The stress applied from the first non-magnetic member **31** may decrease as the distance from the first non-magnetic member **31** increases.

[0056] As shown in FIG. 1, for example, the first magnetic member **21** includes a first position **21a** and a second position **21b**. The second position **21b** is between the first position **21a** and the first non-magnetic member **31**. A first position lattice length along a crossing direction crossing the first direction **D1** at the first position **21a** is different from a second position lattice length along the

crossing direction at the second position **21b**. The crossing direction is, for example, any direction along the X-Y plane (for example, the Y-axis direction). The first position **21a** is far from the first non-magnetic member **31**. The second position **21b** is close to the first non-magnetic member **31**. [0057] For example, in a case where the first magnetic member **21** receives a tensile stress, the second position lattice length is longer than the first position lattice length.

[0058] For example, in a case where the first magnetic member **21** receives a compressive stress, the second position lattice length is shorter than the first position lattice length.

[0059] As described above, the first material includes the first element (one of the first type element and the second type element, for example Ru) and a second element (at least one selected from the group consisting of oxygen and nitrogen). For example, the introduction of the second element into a member including the first element causes the volume of the member to change (for example, increase or decrease). It is preferable that the second element is introduced uniformly in the thickness direction of the first non-magnetic member **31**. This makes it easier for a large stress to be applied from the first non-magnetic member **31** to the first magnetic member **21**.

[0060] As shown in FIG. 1, the first non-magnetic member **31** includes a first non-magnetic portion **31a** and a second non-magnetic portion **31b**. The first non-magnetic portion **31a** includes a first face **31f**. The second non-magnetic portion **31b** includes a second face **31g**. The first face **31f** faces the first magnetic member **21**, for example. The first face **31f** contacts the first magnetic member **21**, for example. The first face **31f** is between the first magnetic member **21** and the second face **31g** in the first direction **D1**. The first non-magnetic portion **31a** is a deep portion. The second non-magnetic portion **31b** is a shallow portion.

[0061] A difference between a concentration of the second element in the first non-magnetic portion **31a** (first concentration) and a concentration of the second element in the second non-magnetic portion **31b** (second concentration) is small. For example, a first ratio of an absolute value of a difference between the first concentration and the second concentration to the first concentration is equal to or less than 0.2.

[0062] For example, the second element (oxygen and/or nitrogen) is introduced without a large difference in concentration between the first non-magnetic portion **31a** (deep portion) and the second non-magnetic portion **31b** (shallow portion). This allows a large stress to be obtained stably.

[0063] In the embodiment, a part of the first magnetic member **21** may include the second element. For example, the concentration of the second element in the first magnetic member **21** may decrease with increasing distance from the first non-magnetic member **31**. As already explained, the first magnetic member **21** includes the first position **21a** and the second position **21b**. The second position **21b** is between the first position **21a** and the first non-magnetic member **31**. The concentration of the second element in the second position **21b** is higher than the concentration of the second element in the first position **21a**. Or, the first position **21a** does not include oxygen.

[0064] In one example of forming the first non-magnetic member **31**, after forming a film including the first element, the film may be treated with a gas including the second element. For example, when the first element is Ru and the second element is oxygen, a set of forming a Ru film with 1 nm thickness and treating with oxygen may be repeated plurality of times. Oxygen is incorporated into the Ru film by treating the Ru film with 1 nm thickness with oxygen. By repeating these sets plurality of times, oxygen is introduced into each of the plurality of Ru films at a stable high concentration. By repeating the above set ten times, the first non-magnetic member **31** with a thickness of about 10 nm is obtained. For example, a low first ratio is easily obtained. Heat treatment may be performed appropriately.

[0065] As described above, the first non-magnetic member **31** may include the second material. In this case, the difference between the concentration of the third element in the first non-magnetic portion **31a** and the concentration of the third element in the second non-magnetic portion **31b** is small. For example, a first ratio of the absolute value of the difference between these concentrations to the concentration of the third element in the first non-magnetic portion **31a** is 0.2

or less.

[0066] The difference between the concentration of the third element in the first non-magnetic portion **31a** and the concentration of the fourth element in the second non-magnetic portion **31b** is small. For example, a first ratio of the absolute value of the difference between these concentrations to the concentration of the fourth element in the first non-magnetic portion **31a** is 0.2 or less.

[0067] In the first non-magnetic member **31**, for example, a film including the third element (at least one selected from the group consisting of Pt, Cu, and Hf) and a film including a fourth element (Al) may be alternately provided, and a heat treatment may be performed.

[0068] In a case where the first non-magnetic member **31** includes Al and Pt, at least one of Pt.sub.2Al.sub.3 and PtAl.sub.2 may be formed by heat treatment at not less than 300° C. and not more than 550° C. The first non-magnetic member **31** may include, for example, an intermetallic compound. Stress occurs due to the formation of the intermetallic compound.

[0069] In a case where the first non-magnetic member **31** includes Al and Cu, a phase transition may occur by heat treatment at not less than 200° C. and not more than 400° C. For example, Al-4Cu may be formed.

[0070] In a case where the first non-magnetic member **31** includes Al and Hf, HfAl.sub.3 may be formed by heat treatment at not less than 350° C. and not more than 650° C. For example, a phase transition occurs from the a phase to the p phase.

[0071] As described above, stress is also generated in a case where the second material described above is applied to the first non-magnetic member **31**. For example, after the film that becomes the first non-magnetic member **31** is formed, a heat treatment is performed at not less than 200° C. and not more than 650° C. The temperature of the heat treatment may be, for example, not less than 300° C. and not more than 400° C. This condition provides good compatibility with the memory manufacturing process, for example. Stress is generated by the heat treatment after the film is formed. The stress is thought to be related to, for example, mutual diffusion of elements, changes in crystal structure, and/or changes in lattice constant due to the heat treatment.

[0072] In a case where a film including the third element and a film including the fourth element are alternately stacked on the first magnetic member **21**, the film including the third element may be in contact with the first magnetic member **21**. The fourth element (Al) is relatively more likely to move (diffuse) than the third element. By having the film including the third element in contact with the first magnetic member **21**, the movement of the fourth element can be suppressed.

[0073] In a case also where the above-mentioned second material is applied to the first non-magnetic member **31**, the configuration described for the first material may be provided.

[0074] In the embodiment, a thickness **t31** (see FIG. 1) of the first non-magnetic member **31** along the first direction **D1** is, for example, not less than 1 nm and not more than 50 nm. By the thickness **t31** being 1 nm or more, for example, a stable stress is easily obtained, and the magnetization of the first magnetic member **21** can be stably controlled. By the thickness **t31** being 50 nm or less, for example, the processing of the element becomes easier, and fine elements can be easily obtained.

[0075] In an embodiment, a thickness **t11** (see FIG. 1) of the first magnetic layer **11** along the first direction **D1** may be, for example, not less than 0.5 nm and not more than 10 nm. The thickness **t15** (see FIG. 1) of the first intermediate layer **15** along the first direction **D1** may be, for example, not less than 0.3 nm and not more than 3 nm.

[0076] As already described, the magnetic device **110** may further include a first conductive member **51**. At least a part of the first conductive member **51** may be in contact with the first magnetic layer **11**. The first conductive member **51** includes, for example, at least one selected from the group consisting of Ta, W, Pt, Hf, Re, Os, Ir, Pd, Cu, Ag, and Au.

[0077] For example, a current flowing through the first conductive member **51** acts on the first magnetic layer **11**. This allows the magnetization of the first magnetic layer **11** (magnetic free layer) to be controlled. For example, the magnetization of the first magnetic layer **11** is controlled

by spin-orbit torque.

[0078] As shown in FIG. 2, the magnetic device **110** may include a controller **70**. The controller **70** is electrically connected to, for example, the first conductive member **51** and the first non-magnetic member **31**.

[0079] As already explained (see FIG. 2), the first conductive member **51** includes a first conductive portion **51a**, a second conductive portion **51b**, and a third conductive portion **51c**. The third conductive portion **51c** is provided between the first conductive portion **51a** and the second conductive portion **51b**. The boundary between these portions may be unclear.

[0080] The first magnetic layer **11** is between the third conductive portion **51c** and the first non-magnetic member **31** in the first direction **D1**. The first magnetic layer **11** overlaps the third conductive portion **51c** in the first direction **D1**. The first magnetic layer **11** does not overlap the first conductive portion **51a** and the second conductive portion **51b** in the first direction **D1**.

[0081] The controller **70** can supply a first current **i1** between the first conductive portion **51a** and the second conductive portion **51b**. The first current **i1** may be oriented from the first conductive portion **51a** to the second conductive portion **51b**, or from the second conductive portion **51b** to the first conductive portion **51a**. The controller **70** can apply a voltage **Val** between the first conductive member **51** and the first non-magnetic member **31**.

[0082] The electrical resistance between the first conductive member **51** and the first non-magnetic member **31** can be changed (controlled) by the direction of the first current **i1** and the voltage **Val** between the first conductive member **51** and the first non-magnetic member **31**. For example, when the voltage **Val** is either negative or positive, the magnetization of the first magnetic layer **11** is likely to change in response to the first current **i1**. For example, when the voltage **Val** is the other of negative and positive, the magnetization of the first magnetic layer **11** is unlikely to change even when the first current **i1** is supplied. With this configuration, the magnetization of the first magnetic layer **11** can be stably controlled. The electrical resistance can be stably controlled.

[0083] FIG. 3 is a schematic cross-sectional view illustrating a magnetic device according to the first embodiment.

[0084] As shown in FIG. 3, a magnetic device **111** according to the embodiment includes the first element portion **10E**. The first element portion **10E** includes a first intermediate member **35**. The configuration of the magnetic device **111** except for this may be similar to the configuration of the magnetic device **110**.

[0085] For example, in the magnetic device **111**, the first element portion **10E** includes the first magnetic layer **11**, the first non-magnetic member **31**, the first magnetic member **21**, the first intermediate layer **15**, and the first intermediate member **35**. The first non-magnetic member **31** is conductive. The direction from the first magnetic layer **11** to the first non-magnetic member **31** is along the first direction **D1**. The first magnetic member **21** is provided between the first magnetic layer **11** and the first non-magnetic member **31**. The first intermediate layer **15** is provided between the first magnetic layer **11** and the first magnetic member **21**, and is non-magnetic. The first intermediate member **35** is provided between the first magnetic member **21** and the first non-magnetic member **31** and is in contact with the first magnetic member **21** and the first non-magnetic member **31**.

[0086] The first non-magnetic member **31** includes at least one of the first material and the second material. The first material includes a first element including one of the first type element and the second type element, and the second element including at least one selected from the group consisting of oxygen and nitrogen. The first type element includes at least one selected from the group consisting of Ru, Ta, Mo, W, Hf, Cr, Cu, Pd, V, Ti, and Zn. The second type element includes at least one selected from the group consisting of Mg and Al. The second material includes the third element including at least one selected from the group consisting of Pt, Cu, and Hf, and the fourth element including Al.

[0087] The first intermediate member **35** includes at least one of a first intermediate material, a

second intermediate material, or a third intermediate material. The first intermediate material includes a fifth element including at least one selected from the group consisting of Mg, Al, Ta, Mo, Nb, Hf, and Ru, and oxygen. The second intermediate material includes a sixth element including at least one selected from the group consisting of B, Si, Ga, and Ti, and nitrogen. The third intermediate material includes at least one selected from the group consisting of W, Re, Os, Ta, Mo, Ir, Ru, and Hf.

[0088] The first intermediate member **35** includes, for example, at least one selected from the group consisting of MgO, Al.sub.2O.sub.3, TaO, MoO.sub.3, NbO, HfO, RuO.sub.2, and Ta.sub.2O.sub.5. Thus, the first intermediate member **35** may include an oxide. In this case, the thickness **t35** (see FIG. **3**) of the first intermediate member **35** is, for example, 0.5 nm or less.

[0089] The first intermediate member **35** includes, for example, at least one selected from the group consisting of BN, SiN, GaN, and TiN. Thus, the first intermediate member **35** may include a nitride. In this case, the thickness **t35** of the first intermediate member **35** is, for example, 1 nm or less.

[0090] The first intermediate member **35** may include, for example, at least one selected from the group consisting of a W region, a Re region, an Os region, a Ta region, a Mo region, an Ir region, a Ru region, and an Hf region. The melting points of these elements are high. These materials are difficult to mix with elements such as Co or Fe provided in the first magnetic member **21**. For example, the third intermediate material is a non-solid-soluble metal with respect to Co or Fe.

[0091] By providing the first intermediate member **35**, it is possible to suppress the movement of elements included in the first non-magnetic member **31** (elements included in the first material or the second material) to the first magnetic member **21**. The first intermediate member **35** is, for example, a diffusion suppression layer.

[0092] By providing the first intermediate member **35**, for example, the first magnetic member **21** becomes stable. Stable characteristics are easily obtained.

[0093] In a case where a film including the third element and a film including the fourth element are alternately stacked in the first intermediate member **35**, the film including the third element may be in contact with the first intermediate member **35**. The fourth element (Al) is relatively more likely to move (diffuse) than the third element. By having the film including the third element in contact with the first intermediate member **35**, the movement of the fourth element can be further suppressed.

[0094] Below, several examples will be described in which stress anisotropy is introduced by an insulating member provided around the first element portion **10E**.

[0095] FIGS. **4A** and **4B** are schematic cross-sectional views illustrating a magnetic device according to the first embodiment.

[0096] As shown in FIGS. **4A** and **4B**, a magnetic device **120** according to the embodiment includes a first insulating member **41** and a second insulating member **42** in addition to the first element portion **10E**. The configuration of the magnetic device **120** except for this may be similar to the configuration of the magnetic device **110**. In this example, the magnetic device **120** includes a first opposing insulating member **41A** and a second opposing insulating member **42A**.

[0097] A direction from the first insulating member **41** to the first element portion **10E** is along the second direction **D2**. The first element portion **10E** is between the first insulating member **41** and the first opposing insulating member **41A** in the second direction **D2**. For example, the first insulating member **41** contacts the first conductive portion **51a**. For example, the first opposing insulating member **41A** contacts the second conductive portion **51b**. The first insulating member **41** contacts at least a part of the first element portion **10E**. The first opposing insulating member **41A** contacts at least a part of the first element portion **10E**.

[0098] As shown in FIG. **4B**, a direction from the second insulating member **42** to the first element portion **10E** is along the third direction **D3**. The first element portion **10E** is between the second insulating member **42** and the second opposing insulating member **42A** in the third direction **D3**. In

this example, the second insulating member **42** contacts at least a part of the first element portion **10E**. The second opposing insulating member **42A** contacts at least a part of the first element portion **10E**.

[0099] The material of the second insulating member **42** is different from the material included in the first insulating member **41**. This induces anisotropy in the stress. For example, anisotropic stress is applied to the first magnetic member **21**. For example, the material of the second opposing insulating member **42A** is different from the material included in the first opposing insulating member **41A**. This causes anisotropic stress to be applied to the first magnetic member **21**.

[0100] The Young's modulus of the material of the second insulating member **42** is different from the Young's modulus of the material included in the first insulating member **41**.

[0101] For example, the Young's modulus of the material of the second insulating member **42** is lower than the Young's modulus of the material included in the first insulating member **41**. In this case, the magnetostriction constant of the first magnetic member **21** may be positive. For example, the lattice length of the first magnetic member **21** along the third direction **D3** is longer than the lattice length of the first magnetic member **21** along the second direction **D2**.

[0102] In another example, the Young's modulus of the material of the second insulating member **42** is higher than the Young's modulus of the material included in the first insulating member **41**. In this case, the magnetostriction constant of the first magnetic member **21** may be negative. For example, the lattice length of the first magnetic member **21** along the third direction **D3** is shorter than the lattice length of the first magnetic member **21** along the second direction **D2**.

[0103] The first insulating member **41** includes, for example, at least one selected from the group consisting of oxygen and nitrogen, and at least one selected from the group consisting of silicon and aluminum. The second insulating member **42** includes, for example, carbon. The second insulating member **42** may include, for example, carbon, at least one selected from the group consisting of oxygen and nitrogen, and at least one selected from the group consisting of silicon and aluminum. The first insulating member **41** does not include, for example, substantially any carbon. Or, the concentration of carbon included in the first insulating member **41** is lower than the concentration of carbon included in the second insulating member **42**.

[0104] The material of the first opposing insulating member **41A** may be the same as the material of the first insulating member **41**. The material of the second opposing insulating member **42A** may be the same as the material of the second insulating member **42**.

[0105] FIGS. **5A** and **5B** are schematic cross-sectional views illustrating the magnetic device according to the first embodiment.

[0106] As shown in FIGS. **5A** and **5B**, a magnetic device **121** according to the embodiment includes the first insulating member **41** and the second insulating member **42** in addition to the first element portion **10E**. The configuration of the magnetic device **121** except for this may be similar to the configuration of the magnetic device **111**.

[0107] In the magnetic device **121**, the direction from the first insulating member **41** to the first element portion **10E** is along the second direction **D2**. The first insulating member **41** is in contact with at least a part of the first element portion **10E**. The direction from the second insulating member **42** to the first element portion **10E** is along the third direction **D3**.

[0108] In the magnetic device **121**, a gap **42g** is provided between the second insulating member **42** and the first element portion **10E**. Such a first insulating member **41** and a second insulating member **42** (gap **42g**) induces anisotropy in the stress. For example, anisotropic stress is applied to the first magnetic member **21**.

[0109] For example, in the second direction **D2**, the first element portion **10E** may be provided between the first insulating member **41** and the first opposing insulating member **41A**. The first opposing insulating member **41A** contacts the first element portion **10E**. In the third direction **D3**, the first element portion **10E** may be provided between the second insulating member **42** and the second opposing insulating member **42A**. A gap **42Ag** is provided between the first element portion

10E and the second opposing insulating member **42A**. Anisotropy is induced in the stress. For example, an anisotropic stress is applied to the first magnetic member **21**.

[0110] In the magnetic device **121**, for example, the magnetostriction constant of the first magnetic member **21** may be positive. For example, the lattice length of the first magnetic member **21** along the third direction **D3** is longer than the lattice length of the first magnetic member **21** along the second direction **D2**.

[0111] In the embodiment, in a case where the magnetostriction constant of the first magnetic member **21** is negative, a gap may be provided between the first insulating member **41** and the first element portion **10E**. For example, the lattice length of the first magnetic member **21** along the third direction **D3** is shorter than the lattice length of the first magnetic member **21** along the second direction **D2**.

[0112] FIGS. **6A** and **6B** are schematic cross-sectional views illustrating the magnetic device according to the first embodiment.

[0113] As shown in FIGS. **6A** and **6B**, a magnetic device **122** according to the embodiment includes the first insulating member **41** and the second insulating member **42** in addition to the first element portion **10E**. The configuration of the magnetic device **122** except for this may be similar to the configuration of the magnetic device **111**. For example, the first element portion **10E** includes the first intermediate member **35**. The material of the second insulating member **42** is different from the material of the first insulating member **41**.

[0114] FIGS. **7A** and **7B** are schematic cross-sectional views illustrating the magnetic device according to the first embodiment.

[0115] As shown in FIG. **7A** and FIG. **7B**, a magnetic device **123** according to the embodiment includes the first insulating member **41** and the second insulating member **42** in addition to the first element portion **10E**. The configuration of the magnetic device **123** except for this may be similar to the configuration of the magnetic device **111**. For example, the first element portion **10E** includes the first intermediate member **35**. The first insulating member **41** contacts the first element portion **10E**. A gap **42g** is provided between the second insulating member **42** and the first element portion **10E**.

[0116] In the magnetic device **122** and the magnetic device **123**, anisotropy is induced in the stress. For example, anisotropic stress is applied to the first magnetic member **21**.

Second Embodiment

[0117] FIG. **8** is a schematic cross-sectional view illustrating a magnetic device according to a second embodiment.

[0118] As shown in FIG. **8**, a magnetic device **130** according to the embodiment further includes a second non-magnetic member **32** in addition to the first element portion **10E**. The configuration of the magnetic device **130** except for this may be similar to that of the magnetic device **110**.

[0119] The first magnetic layer **11** is between the second non-magnetic member **32** and at least a part of the first conductive member **51** in the first direction **D1**. The second non-magnetic member **32** may be in contact with the first conductive member **51**. The second non-magnetic member **32** includes at least one of the third material and the fourth material. The third material includes one of a third element and a fourth element, and at least one selected from the group consisting of oxygen and nitrogen. The third element includes at least one selected from the group consisting of Ru, Ta, Mo, W, Hf, Cr, Cu, Pd, V, Ti, and Zn. The fourth element includes at least one selected from the group consisting of Mg and Al. The fourth material includes at least one selected from the group consisting of Pt, Cu, and Hf, and Al.

[0120] The second non-magnetic member **32** may, for example, apply stress to the first conductive member **51** and the first element portion **10E**. The second non-magnetic member **32** may be conductive or non-conductive. The second non-magnetic member **32** provides a magnetic device capable of more stable operation. The second non-magnetic member **32** may be applied to any of the magnetic devices described in relation to the first embodiment.

Third Embodiment

[0121] FIG. 9 is a schematic cross-sectional view illustrating a magnetic device according to a third embodiment.

[0122] As shown in FIG. 9, a magnetic device 140 according to the embodiment includes a plurality of first element portions 10E. The magnetic device 140 can be used, for example, as a memory circuit. The magnetic device 120 can be used, for example, as a logic circuit. The number of the plurality of first element portions 10E is arbitrary.

[0123] For example, the first conductive member 51 may further include a fourth conductive portion 51d and a fifth conductive portion 51e. The second conductive portion 51b is between the first conductive portion 51a and the fourth conductive portion 51d. The fifth conductive portion 51e is between the second conductive portion 51b and the fourth conductive portion 51d. One of the plurality of first element portions 10E is provided in the third conductive portion 51c. Another one of the plurality of first element portions 10E is provided in the fifth conductive portion 51e.

[0124] The configuration of the magnetic device 140 may be applied to any magnetic device according to the first or second embodiment.

[0125] The embodiment may include the following configurations:

(Configuration 1)

[0126] A magnetic device, comprising: [0127] a first element portion, the first element portion including: [0128] a first magnetic layer; [0129] a non-magnetic member being conductive, a direction from the first magnetic layer to the first non-magnetic member being along a first direction; [0130] a first magnetic member provided between the first magnetic layer and the first non-magnetic member; and [0131] a first intermediate layer being non-magnetic, the first intermediate layer being provided between the first magnetic layer and the first magnetic member, [0132] the first material including a first element and a second element, the first element including one of a first type element and a second type element, the second element including at least one element selected from the group consisting of oxygen and nitrogen, [0133] the first type element including at least one element selected from the group consisting of Ru, Ta, Mo, W, Hf, Cr, Cu, Pd, V, Ti, and Zn, [0134] the second type element including at least one selected from the group consisting of Mg and Al, and [0135] the second material including a third element and a fourth element, the third element including at least one selected from the group consisting of Pt, Cu, and Hf, the fourth element including Al.

(Configuration 2)

[0136] A magnetic device, comprising: [0137] a first element portion, the first element portion including: [0138] a first magnetic layer; [0139] a non-magnetic member being conductive, a direction from the first magnetic layer to the first non-magnetic member being along a first direction; [0140] a first magnetic member provided between the first magnetic layer and the first non-magnetic member; [0141] a first intermediate layer being non-magnetic, the first intermediate layer being provided between the first magnetic layer and the first magnetic member, and [0142] a first intermediate member provided between the first magnetic member and the first non-magnetic member and being in contact with the first magnetic member and the first non-magnetic member; [0143] the first material including a first element and a second element, the first element including one of a first type element and a second type element, the second element including at least one element selected from the group consisting of oxygen and nitrogen, [0144] the first type element including at least one element selected from the group consisting of Ru, Ta, Mo, W, Hf, Cr, Cu, Pd, V, Ti, and Zn, [0145] the second type element including at least one selected from the group consisting of Mg and Al, [0146] the second material including a third element and a fourth element, the third element including at least one selected from the group consisting of Pt, Cu, and Hf, the fourth element including Al, [0147] the first intermediate member including at least one of a first intermediate material, a second intermediate material, or a third intermediate material, [0148] the first intermediate material including oxygen and a fifth element including at least one selected from

the group consisting of Mg, Al, Ta, Mo, Nb, Hf, and Ru, [0149] the second intermediate material including nitrogen and a sixth element including at least one selected from the group consisting of B, Si, Ga, and T, and [0150] the third intermediate material including at least one selected from the group consisting of W, Re, Os, Ta, Mo, Ir, Ru, and Hf.

(Configuration 3)

[0151] The magnetic device according to Configuration 1 or 2, further comprising: [0152] a first conductive member, [0153] the first conductive member including a first conductive portion, a second conductive portion, and a third conductive portion, [0154] the third conductive portion being provided between the first conductive portion and the second conductive portion in a second direction crossing the first direction, [0155] the first magnetic layer being between the third conductive portion and the first non-magnetic member in the first direction, [0156] the first conductive member including at least one selected from the group consisting of Ta, W, Pt, Hf, Re, Os, Ir, Pd, Cu, Ag, and Au.

(Configuration 4)

[0157] The magnetic device according to Configuration 3, further comprising: [0158] a first insulating member and a second insulating member, [0159] a direction from the first insulating member to the first element portion being along the second direction, [0160] the first insulating member being in contact with at least a part of the first element portion, [0161] a direction from the second insulating member to the first element portion being along a third direction crossing a plane including the first direction and the second direction, [0162] the second insulating member being in contact with at least a part of the first element portion, and [0163] a material of the second insulating member being different from a material included in the first insulating member.

(Configuration 5)

[0164] The magnetic device according to Configuration 4, wherein [0165] a Young's modulus of the material of the second insulating member is different from a Young's modulus of the material included in the first insulating member.

(Configuration 6)

[0166] The magnetic device according to Configuration 3, further comprising: [0167] a first insulating member and a second insulating member, [0168] a direction from the first insulating member to the first element portion being along the second direction, [0169] the first insulating member being in contact with at least a part of the first element portion, [0170] a direction from the second insulating member to the first element portion being along a third direction crossing a plane including the first direction and the second direction, and [0171] a gap being provided between the second insulating member and the first element portion.

(Configuration 7)

[0172] The magnetic device according to Configuration 5, wherein [0173] a magnetostriction constant of the first magnetic member is positive, and [0174] the Young's modulus of the material of the second insulating member is lower than the Young's modulus of the material included in the first insulating member.

(Configuration 8)

[0175] The magnetic device according to Configuration 7, wherein a magnetostriction constant of the first magnetic member is positive.

(Configuration 9)

[0176] The magnetic device according to Configuration 1 or 2, wherein [0177] a length of the first non-magnetic member along a third direction is longer than a length of the first non-magnetic member along a second direction, [0178] the second direction crosses the first direction, and [0179] the third direction crosses a plane including the first direction and the second direction.

(Configuration 10)

[0180] The magnetic device according to Configuration 9, further comprising: [0181] a first conductive member, [0182] the first conductive member including a first conductive portion, a

second conductive portion, and a third conductive portion, [0183] the third conductive portion being provided between the first conductive portion and the second conductive portion in the second direction, [0184] the first magnetic layer being between the third conductive portion and the first non-magnetic member in the first direction, and [0185] the first conductive member including at least one selected from the group consisting of Ta, W, Pt, Hf, Re, Os, Ir, Pd, Cu, Ag, and Au.

(Configuration 11)

[0186] The magnetic device according to Configuration 9 or 10, wherein a magnetostriction constant of the first magnetic member is positive.

(Configuration 12)

[0187] The magnetic device according to any one of Configurations 4-11, wherein [0188] a third direction lattice length along the third direction of at least a part of the first magnetic member is longer than a second direction lattice length along the second direction of at least a part of the first magnetic member.

(Configuration 13)

[0189] The magnetic device according to any one of Configurations 1 to 12, wherein [0190] the first magnetic member includes a plurality of first magnetic films and a plurality of first non-magnetic films, and [0191] in the first direction, one of the plurality of first non-magnetic films is between one of the plurality of first magnetic films and another one of the plurality of first magnetic films.

(Configuration 14)

[0192] The magnetic device according to Configuration 13, wherein [0193] the first magnetic member satisfies a first condition or a second condition, [0194] in the first condition, the one of the plurality of first non-magnetic films includes Ru, and a thickness of the one of the plurality of first non-magnetic films along the first direction is not less than 0.2 nm and not more than 2 nm, and [0195] in the second condition, the one of the plurality of first non-magnetic films includes Ir, and the thickness of the one of the plurality of first non-magnetic films along the first direction is not less than 0.2 nm and not more than 2 nm.

(Configuration 15)

[0196] The magnetic device according to any one of Configurations 1-14, wherein [0197] the first non-magnetic member includes a first non-magnetic portion including a first face and a second non-magnetic portion including a second face, [0198] the first face is between the first magnetic member and the second face in the first direction, [0199] a first ratio of an absolute value of a difference between a first concentration of the second element in the first non-magnetic portion and a second concentration of the second element in the second non-magnetic portion to the first concentration is 0.2 or less.

(Configuration 16)

[0200] The magnetic device according to any one of Configurations 1 to 15, wherein [0201] the first magnetic member includes a first position and a second position, [0202] the second position is between the first position and the first non-magnetic member, [0203] a concentration of the second element at the second position is higher than a concentration of the second element at the first position, or the first position does not include oxygen.

(Configuration 17)

[0204] The magnetic device according to any one of Configurations 3-8, further comprising: [0205] a controller, [0206] the controller being configured to supply a first current between the first conductive portion and the second conductive portion, [0207] an electrical resistance between the first conductive member and the first non-magnetic member being configured to be changed depending on a direction of the first current and a voltage between the first conductive member and the first non-magnetic member.

(Configuration 18)

[0208] The magnetic device according to any one of Configurations 3-8, wherein [0209] the first

magnetic member includes a first position and a second position, [0210] the second position is between the first position and the first non-magnetic member, [0211] a first position lattice length along a cross direction crossing the first direction at the first position being different from a second position lattice length along the cross direction at the second position.

(Configuration 19)

[0212] The magnetic device according to Configuration 18, wherein [0213] the second position lattice length is longer than the first position lattice length.

(Configuration 20)

[0214] The magnetic device according to any one of configurations 3-8, further comprising: [0215] a second non-magnetic member, [0216] the first magnetic layer being between the second non-magnetic member and at least a part of the first conductive member in the first direction, and [0217] the second non-magnetic member being in contact with the first conductive member, [0218] the second non-magnetic member including at least one of a third material or a fourth material, [0219] the third material including one of a third type element and a fourth type element, and at least one selected from the group consisting of oxygen and nitrogen, [0220] the third type element including at least one selected from the group consisting of Ru, Ta, Mo, W, Hf, Cr, Cu, Pd, V, Ti, and Zn, [0221] the fourth type element including at least one selected from the group consisting of Mg and Al, and [0222] the fourth material including at least one selected from the group consisting of Pt, Cu, and Hf, and Al.

[0223] According to the embodiment, a magnetic device capable of stable operation can be provided.

[0224] Hereinabove, exemplary embodiments of the invention are described with reference to specific examples. However, the embodiments of the invention are not limited to these specific examples. For example, one skilled in the art may similarly practice the invention by appropriately selecting specific configurations of components included in the magnetic devices such as element portions, magnetic layers, non-magnetic members, magnetic members, intermediate layers, intermediate members, conductive members, insulating members, etc., from known art. Such practice is included in the scope of the invention to the extent that similar effects thereto are obtained.

[0225] Further, any two or more components of the specific examples may be combined within the extent of technical feasibility and are included in the scope of the invention to the extent that the purport of the invention is included.

[0226] Moreover, all magnetic devices practicable by an appropriate design modification by one skilled in the art based on the magnetic devices described above as embodiments of the invention also are within the scope of the invention to the extent that the purport of the invention is included.

[0227] Various other variations and modifications can be conceived by those skilled in the art within the spirit of the invention, and it is understood that such variations and modifications are also encompassed within the scope of the invention.

Claims

1. A magnetic device, comprising: a first element portion, the first element portion including: a first magnetic layer; a non-magnetic member being conductive, a direction from the first magnetic layer to the first non-magnetic member being along a first direction; a first magnetic member provided between the first magnetic layer and the first non-magnetic member; and a first intermediate layer being non-magnetic, the first intermediate layer being provided between the first magnetic layer and the first magnetic member, the first material including a first element and a second element, the first element including one of a first type element and a second type element, the second element including at least one element selected from the group consisting of oxygen and nitrogen, the first type element including at least one element selected from the group consisting of Ru, Ta, Mo, W,

Hf, Cr, Cu, Pd, V, Ti, and Zn, the second type element including at least one selected from the group consisting of Mg and Al, and the second material including a third element and a fourth element, the third element including at least one selected from the group consisting of Pt, Cu, and Hf, the fourth element including Al.

2. A magnetic device, comprising: a first element portion, the first element portion including: a first magnetic layer; a non-magnetic member being conductive, a direction from the first magnetic layer to the first non-magnetic member being along a first direction; a first magnetic member provided between the first magnetic layer and the first non-magnetic member; a first intermediate layer being non-magnetic, the first intermediate layer being provided between the first magnetic layer and the first magnetic member, and a first intermediate member provided between the first magnetic member and the first non-magnetic member and being in contact with the first magnetic member and the first non-magnetic member; the first material including a first element and a second element, the first element including one of a first type element and a second type element, the second element including at least one element selected from the group consisting of oxygen and nitrogen, the first type element including at least one element selected from the group consisting of Ru, Ta, Mo, W, Hf, Cr, Cu, Pd, V, Ti, and Zn, the second type element including at least one selected from the group consisting of Mg and Al, the second material including a third element and a fourth element, the third element including at least one selected from the group consisting of Pt, Cu, and Hf, the fourth element including Al, the first intermediate member including at least one of a first intermediate material, a second intermediate material, or a third intermediate material, the first intermediate material including oxygen and a fifth element including at least one selected from the group consisting of Mg, Al, Ta, Mo, Nb, Hf, and Ru, the second intermediate material including nitrogen and a sixth element including at least one selected from the group consisting of B, Si, Ga, and T, and the third intermediate material including at least one selected from the group consisting of W, Re, Os, Ta, Mo, Ir, Ru, and Hf.

3. The magnetic device according to claim 1, further comprising: a first conductive member, the first conductive member including a first conductive portion, a second conductive portion, and a third conductive portion, the third conductive portion being provided between the first conductive portion and the second conductive portion in a second direction crossing the first direction, the first magnetic layer being between the third conductive portion and the first non-magnetic member in the first direction, the first conductive member including at least one selected from the group consisting of Ta, W, Pt, Hf, Re, Os, Ir, Pd, Cu, Ag, and Au.

4. The magnetic device according to claim 3, further comprising: a first insulating member and a second insulating member, a direction from the first insulating member to the first element portion being along the second direction, the first insulating member being in contact with at least a part of the first element portion, a direction from the second insulating member to the first element portion being along a third direction crossing a plane including the first direction and the second direction, the second insulating member being in contact with at least a part of the first element portion, and a material of the second insulating member being different from a material included in the first insulating member.

5. The magnetic device according to claim 4, wherein a Young's modulus of the material of the second insulating member is different from a Young's modulus of the material included in the first insulating member.

6. The magnetic device according to claim 3, further comprising: a first insulating member and a second insulating member, a direction from the first insulating member to the first element portion being along the second direction, the first insulating member being in contact with at least a part of the first element portion, a direction from the second insulating member to the first element portion being along a third direction crossing a plane including the first direction and the second direction, and a gap being provided between the second insulating member and the first element portion.

7. The magnetic device according to claim 5, wherein a magnetostriction constant of the first

magnetic member is positive, and the Young's modulus of the material of the second insulating member is lower than the Young's modulus of the material included in the first insulating member.

8. The magnetic device according to claim 7, wherein a magnetostriction constant of the first magnetic member is positive.

9. The magnetic device according to claim 1, wherein a length of the first non-magnetic member along a third direction is longer than a length of the first non-magnetic member along a second direction, the second direction crosses the first direction, and the third direction crosses a plane including the first direction and the second direction.

10. The magnetic device according to claim 9, further comprising: a first conductive member, the first conductive member including a first conductive portion, a second conductive portion, and a third conductive portion, the third conductive portion being provided between the first conductive portion and the second conductive portion in the second direction, the first magnetic layer being between the third conductive portion and the first non-magnetic member in the first direction, and the first conductive member including at least one selected from the group consisting of Ta, W, Pt, Hf, Re, Os, Ir, Pd, Cu, Ag, and Au.

11. The magnetic device according to claim 9, wherein a magnetostriction constant of the first magnetic member is positive.

12. The magnetic device according to claim 4, wherein a third direction lattice length along the third direction of at least a part of the first magnetic member is longer than a second direction lattice length along the second direction of at least a part of the first magnetic member.

13. The magnetic device according to claim 1, wherein the first magnetic member includes a plurality of first magnetic films and a plurality of first non-magnetic films, and in the first direction, one of the plurality of first non-magnetic films is between one of the plurality of first magnetic films and another one of the plurality of first magnetic films.

14. The magnetic device according to claim 13, wherein the first magnetic member satisfies a first condition or a second condition, in the first condition, the one of the plurality of first non-magnetic films includes Ru, and a thickness of the one of the plurality of first non-magnetic films along the first direction is not less than 0.2 nm and not more than 2 nm, and in the second condition, the one of the plurality of first non-magnetic films includes Ir, and the thickness of the one of the plurality of first non-magnetic films along the first direction is not less than 0.2 nm and not more than 2 nm.

15. The magnetic device according to claim 1, wherein the first non-magnetic member includes a first non-magnetic portion including a first face and a second non-magnetic portion including a second face, the first face is between the first magnetic member and the second face in the first direction, a first ratio of an absolute value of a difference between a first concentration of the second element in the first non-magnetic portion and a second concentration of the second element in the second non-magnetic portion to the first concentration is 0.2 or less.

16. The magnetic device according to claim 1, wherein the first magnetic member includes a first position and a second position, the second position is between the first position and the first non-magnetic member, a concentration of the second element at the second position is higher than a concentration of the second element at the first position, or the first position does not include oxygen.

17. The magnetic device according to claim 3, further comprising: a controller, the controller being configured to supply a first current between the first conductive portion and the second conductive portion, an electrical resistance between the first conductive member and the first non-magnetic member being configured to be changed depending on a direction of the first current and a voltage between the first conductive member and the first non-magnetic member.

18. The magnetic device according to claim 3, wherein the first magnetic member includes a first position and a second position, the second position is between the first position and the first non-magnetic member, a first position lattice length along a cross direction crossing the first direction at the first position being different from a second position lattice length along the cross direction at

the second position.

19. The magnetic device according to claim 18, wherein the second position lattice length is longer than the first position lattice length.

20. The magnetic device according to claim 3, further comprising: a second non-magnetic member, the first magnetic layer being between the second non-magnetic member and at least a part of the first conductive member in the first direction, and the second non-magnetic member being in contact with the first conductive member, the second non-magnetic member including at least one of a third material or a fourth material, the third material including one of a third type element and a fourth type element, and at least one selected from the group consisting of oxygen and nitrogen, the third type element including at least one selected from the group consisting of Ru, Ta, Mo, W, Hf, Cr, Cu, Pd, V, Ti, and Zn, the fourth type element including at least one selected from the group consisting of Mg and Al, and the fourth material including at least one selected from the group consisting of Pt, Cu, and Hf, and Al.
